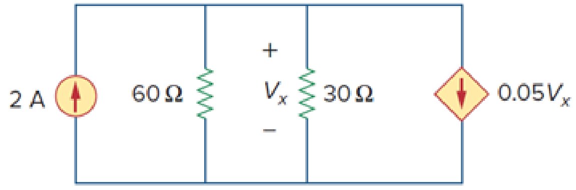
Chapter 3, Problem 7P

Problem

Apply nodal analysis to solve for *Vx* in the circuit of Fig. 3.56.

Figure 3.56:



Step-by-step solution

Step 1 of 2

Refer to Figure 3.56 in the text book.

Assume a node at the resistances $60\ \Omega$ and $30\ \Omega$.

The 2 A current enters into the node.

The current through the $60\ \Omega$ and $30\ \Omega$ resistances, and $0.05V_x$ dependent current source leaves the node.

Step 2 of 2

Apply Kirchhoff's current law for the circuit; the sum of the currents entering the node is equal to the sum of the currents leaving the node.

$$\frac{V_x}{30}+\frac{V_x}{60}+0.05V_x-2=0$$
$$\frac{2V_x+V_x+3V_x-120}{60}=0$$
$$2V_x+V_x+3V_x-120=0$$
$$6V_x=120$$
$$V_x=\frac{120}{6}$$
$$=20\text{ V}$$

Therefore, the value of the voltage V_x is, $\boxed{20\text{ V}}$.